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```
clear; clc; close all;
```

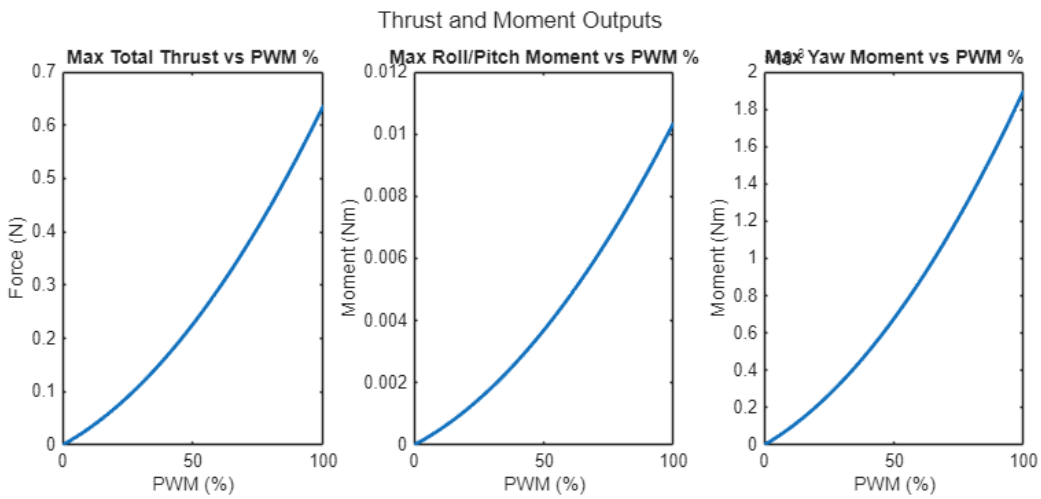
Quadcopter Parameters

```
A_thrust = 0.091492681;
B_thrust = 0.067673604;
L = 0.046;
C_tau = 0.005964552; % torque
d = L * (sqrt(2)/2); % distance between motor and drone COM

% Physical Force Limits per motor (Based on ratio 0.0 to 1.0)
F_min = 0;
F_max = A_thrust*(1)^2 + B_thrust*(1);
```

Task 5: Visualize Thrust and Moments vs PWM Percentage

```
pwm_pct_plot = linspace(0, 100, 100);
pwm_ratio_plot = pwm_pct_plot / 100;

% Nonlinear forces
F_plot = A_thrust * (pwm_ratio_plot.^2) + B_thrust * pwm_ratio_plot;

figure('Position', [100, 100, 1000, 400]);

% Plot Total Thrust
subplot(1,3,1);
plot(pwm_pct_plot, 4 * F_plot, 'LineWidth', 2);
title('Max Total Thrust vs PWM %');
xlabel('PWM (%)'); ylabel('Force (N)');
```

```
% Plot Roll/Pitch Moment
subplot(1,3,2);
plot(pwm_pct_plot, 2 * d * F_plot, 'LineWidth', 2);
title('Max Roll/Pitch Moment vs PWM %');
xlabel('PWM (%)'); ylabel('Moment (Nm)');

% Plot Yaw Moment
subplot(1,3,3);
plot(pwm_pct_plot, 2 * C_tau * F_plot, 'LineWidth', 2);
title('Max Yaw Moment vs PWM %');
xlabel('PWM (%)'); ylabel('Moment (Nm)');

sgtitle('Thrust and Moment Outputs');
```



Test the Mixer (Tasks 1, 2, 3)

```
% Let's command hover thrust + an aggressive roll command
Fc_cmd = 0.323; % ~Hover weight
Lc_cmd = 0.8; % Aggressive roll
Mc_cmd = 0.8;
Nc_cmd = 0;

[pwm_outputs, F_actual] = quad_mixer(Fc_cmd, Lc_cmd, Mc_cmd, Nc_cmd,
A_thrust, B_thrust, d, C_tau, F_min, F_max);

disp('--- Commanded vs Actual ---');
fprintf('Commanded Forces: [%.4f, %.4f, %.4f, %.4f] N\n', Fc_cmd, Lc_cmd,
Mc_cmd, Nc_cmd);
fprintf('Required Motor PWMs (%%): [%.1f%%, %.1f%%, %.1f%%, %.1f%%]\n',
pwm_outputs(1), pwm_outputs(2), pwm_outputs(3), pwm_outputs(4));
```

The Mixer Function

```
function [pwm_pct, F_des] = quad_mixer(Fc, Lc, Mc, Nc, A, B, d, C_tau,
F_min, F_max)
```

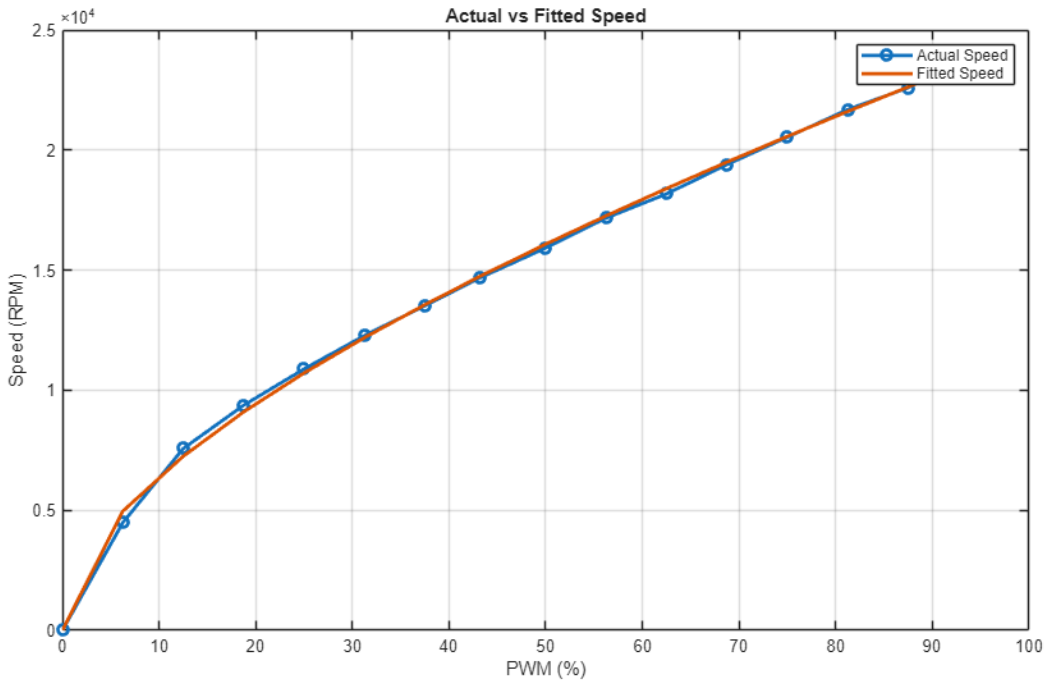
```
    % 1. Linear Mapping Matrix
    M_mix = [1,      1,      1,      1;
            -d,     -d,      d,      d;
             d,     -d,     -d,      d;
             C_tau, -C_tau,  C_tau, -C_tau];

    % Calculate desired forces
    U_cmd = [Fc; Lc; Mc; Nc];
    F_des = M_mix \ U_cmd;

    % 2 & 3. Actuator Saturation & Infeasible Command Handling
    % Strategy: Shift forces to preserve moments, scale if impossible.

    if max(F_des) > F_max
        % Shift down (Sacrifice altitude to keep attitude)
        shift = max(F_des) - F_max;
        F_des = F_des - shift;
    end

    if min(F_des) < F_min
        % Shift up
        shift = F_min - min(F_des);
        F_des = F_des + shift;
    end

    % If the spread is wider than physical limits, we must scale moments down
    if (max(F_des) - min(F_des)) > (F_max - F_min)
        mid_req = mean(F_des);
        mid_phys = (F_max + F_min) / 2;
        scale = (F_max - F_min) / (max(F_des) - min(F_des));

        % Center and scale
        F_des = mid_phys + scale * (F_des - mid_req);
    end

    % Final safety clamp against floating point errors
    F_des = max(min(F_des, F_max), F_min);

    % 4. Nonlinear Inversion (Force to PWM Ratio)
    % Solve: A*x^2 + B*x - F_des = 0 for positive root
    pwm_ratio = (-B + sqrt(B^2 + 4*A.*F_des)) ./ (2*A);

    % 5. Convert to valid percentage output
    pwm_pct = pwm_ratio * 100;
end
```

```
--- Commanded vs Actual ---
Commanded Forces: [0.3230, 0.8000, 0.8000, 0.0000] N
Required Motor PWMs (%): [63.3%, 0.0%, 63.3%, 100.0%]
```

fitting data

```
data = [0.24 0.0 4.01 0 0;
        0.37 1.6 3.98 6.25 4485;
        0.56 4.8 3.95 12.5 7570;
        0.75 7.9 3.92 18.75 9374;
        0.94 10.9 3.88 25 10885;
        1.15 13.9 3.84 31.25 12277;
        1.37 17.3 3.80 37.5 13522;
        1.59 21.0 3.76 43.25 14691;
        1.83 24.4 3.71 50 15924;
        2.11 28.6 3.67 56.25 17174;
        2.39 32.8 3.65 62.5 18179;
        2.71 37.3 3.62 68.75 19397;
        3.06 41.7 3.56 75 20539;
        3.46 46.0 3.48 81.25 21692;
        3.88 51.9 3.40 87.5 22598;
        4.44 57.9 3.30 93.75 23882];

thrust_data = data(:,2);
pwm_data = data(:, 4);
speed_data = data(:, 5);


ft = fittype('a*sqrt(x) + b*x', 'independent', 'x', 'dependent', 'y');

[cf_2, gof] = fit(pwm_data, speed_data, ft);

disp(cf_2)

predicted_speed = cf_2(pwm_data);

% Plot the predicted speed against actual speed for comparison
figure;
plot(pwm_data, speed_data, '-o', 'DisplayName', 'Actual Speed',
'LineWidth',2);
hold on;
plot(pwm_data, predicted_speed, '-', 'DisplayName', 'Fitted Speed',
'LineWidth',2);
xlabel('PWM (%)');
ylabel('Speed (RPM)');
title('Actual vs Fitted Speed');
legend show;
grid on;
hold off;
```

```
Warning: Start point not provided, choosing random start point.
     General model:
     cf_2(x) = a*sqrt(x) + b*x
     Coefficients (with 95% confidence bounds):
       a =        1826  (1744, 1909)
       b =        63.4  (53.11, 73.69)
```



rebuild new model

```
% constant params
pwm_hover_value = 0.6408; %% value derived from previous assignment.


A_thrust = 0.091492681;
B_thrust = 0.067673604;
L = 0.046;
drone_mass_kg = 0.033;

J = [16.571710 0.830806 0.718277;
      0.830806 16.655602 1.800197;
      0.718277 1.800197 29.261652].*10e-6;

Ts = 0.01; % SAMPLING TIME MUST CHANGE

num_d = 0.0304596;
den_d = [1 , -0.9695404];

sys_discrete = tf(num_d, den_d, Ts, 'Variable', 'z^-1');

sys_continuous = d2c(sys_discrete, 'zoh') % zero order hold

[A_act, B_act, C_act, D_act] = ssdata(sys_continuous); % continuous state
space for the motor dynamics.
```

```
%states (16now)
syms pn pe pd u v w phi theta psi p q r real

%motor state
syms pwm_act_1 pwm_act_2 pwm_act_3 pwm_act_4 real

% motor command input
syms pwm_cmd_1 pwm_cmd_2 pwm_cmd_3 pwm_cmd_4 real

state_vector_2 = [pn; pe; pd; u; v; w; phi; theta; psi; p; q; r; pwm_act_1;
pwm_act_2; pwm_act_3; pwm_act_4];
input_vector_2 = [pwm_cmd_1; pwm_cmd_2; pwm_cmd_3; pwm_cmd_4];




% forces
F1 = A_thrust * pwm_act_1^2 + B_thrust*pwm_act_1;
F2 = A_thrust * pwm_act_2^2 + B_thrust*pwm_act_2;
F3 = A_thrust * pwm_act_3^2 + B_thrust*pwm_act_3;
F4 = A_thrust * pwm_act_4^2 + B_thrust*pwm_act_4;
F_total_body = [0; 0; -(F1 + F2 + F3 + F4)]; % total force in body frame NED

%motor torques
T1 = 0.005964552 * F1;
T2 = 0.005964552 * F2;
T3 = 0.005964552 * F3;
T4 = 0.005964552 * F4;

% motor dynamics
pwm_act1_dot = A_act * pwm_act_1 + B_act * pwm_cmd_1;
pwm_act2_dot = A_act * pwm_act_2 + B_act * pwm_cmd_2;
pwm_act3_dot = A_act * pwm_act_3 + B_act * pwm_cmd_3;
pwm_act4_dot = A_act * pwm_act_4 + B_act * pwm_cmd_4;

% moment vector torques
l = L*(sqrt(2)/2)*(F3 + F4 - F1 - F2);
m = L*(sqrt(2)/2)*(F1 + F4 - F3 - F2);
n = T1 + T3 - T2 - T4;
moments = [l; m; n];




% all rotations
Rx = [1 0 0;
    0 cos(phi) -sin(phi);
    0 sin(phi) cos(phi)];

Ry = [cos(theta) 0 sin(theta);
    0 1 0;
    -sin(theta) 0 cos(theta)];

Rz = [cos(psi) -sin(psi) 0;
```

```
       sin(psi) cos(psi) 0;
       0 0 1];

R_body_to_inertial = Rz * Ry * Rx; % rotation Matrix (Body to Inertial - ZYX
Sequence)


p_dot = R_body_to_inertial * [u; v; w]; % p_dot states 1-3



W = [1, sin(phi)*tan(theta), cos(phi)*tan(theta);
     0, cos(phi),            -sin(phi);
     0, sin(phi)/cos(theta), cos(phi)/cos(theta)];
euler_dot = W * [p; q; r]; % angular dynamics states 7-9

% Gravity rotated to Body Frame
g_inertial = [0; 0; 9.81];
g_body = R_body_to_inertial' * g_inertial;

omega = [p; q; r];
v_body = [u; v; w];


% aroo dynamics
K_aeroo = [-10.2506 -0.3177 -0.4332;
            -0.3177 -10.2506 -0.4332;
            -7.7050 -7.7050 -7.5530].*10e-7;

% these in rev/s
n1 = (1826*sqrt(pwm_act_1) + 63.4*pwm_act_1)/60;
n2 = (1826*sqrt(pwm_act_2) + 63.4*pwm_act_2)/60;
n3 = (1826*sqrt(pwm_act_3) + 63.4*pwm_act_3)/60;
n4 = (1826*sqrt(pwm_act_4) + 63.4*pwm_act_4)/60;


theta_dot_sigma = 2*pi*(abs(n1) + abs(n2) + abs(n3) + abs(n4));

f_aeroo = K_aeroo * theta_dot_sigma * (R_body_to_inertial*v_body); % not
sure if this should be rotated...

v_dot = (1/drone_mass_kg) * (F_total_body + f_aeroo + drone_mass_kg*g_body)
- cross(omega, v_body); % dynamics for states 4-6


omega_dot = J \ (moments - cross(omega, J * omega)); % rigid body calc


f_x_2 = [p_dot;
       v_dot;
       euler_dot;
       omega_dot;
       pwm_act1_dot;
       pwm_act2_dot;
```

```
       pwm_act3_dot;
       pwm_act4_dot];
```

```
sys_continuous =

  0.03046 s + 3.093
  -----------------
      s + 3.093

Continuous-time transfer function.
```

Linearizing the 16-state model.

```
% linearize
A_sym_2 = jacobian(f_x_2, state_vector_2);
B_sym_2 = jacobian(f_x_2, input_vector_2);

% hover state and hover inputs
x0_2 = zeros(16, 1);
x0_2(13:16) = pwm_hover_value;
u0_2 = [pwm_hover_value; pwm_hover_value; pwm_hover_value; pwm_hover_value];

% substitute into linearized system
A_2 = double(subs(A_sym_2, [state_vector_2; input_vector_2], [x0_2; u0_2]))
B_2 = double(subs(B_sym_2, [state_vector_2; input_vector_2], [x0_2; u0_2]))

% start making C and D, which will do a bunch of conversion stuff.
r2d = 180/pi;

C_2 = zeros(16, 16);

% Positions \
C_2(1, 2) = 1;
C_2(2, 1) = 1;
C_2(3, 3) = -1;

% Velocities
C_2(4, 5) = 1;
C_2(5, 4) = 1;
C_2(6, 6) = -1;

% Euler Angles (NED Rads -> ENU Degrees)
C_2(7, 8) = r2d;
C_2(8, 7) = r2d;
C_2(9, 9) = -r2d;

% Angular Rates
% negating state 11 again.
C_2(10, 11) = r2d;
C_2(11, 10) = -r2d;
C_2(12, 12) = -r2d;
```

```
% Motor dynamics
C_2(13, 13) = C_act;
C_2(14, 14) = C_act;
C_2(15, 15) = C_act;
C_2(16, 16) = C_act

% D Matrix is zeros except for the dynamics of the motors:
D_2 = zeros(16, 4);
D_2(13, 1) = D_act;
D_2(14, 2) = D_act;
D_2(15, 3) = D_act;
D_2(16, 4) = D_act
```

```
A_2 =

  Columns 1 through 7

         0         0         0    1.0000         0         0         0
         0         0         0         0    1.0000         0         0
         0         0         0         0         0    1.0000         0
         0         0         0   -0.1955   -0.0061   -0.0083         0
         0         0         0   -0.0061   -0.1955   -0.0083    9.8100
         0         0         0   -0.1469   -0.1469   -0.1440         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0

  Columns 8 through 14

         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
   -9.8100         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0   -5.6040   -5.6040
         0         0    1.0000         0         0         0         0
         0         0         0    1.0000         0         0         0
         0         0         0         0    1.0000         0         0
         0         0         0         0         0  -38.2950  -34.5432
         0         0         0         0         0   37.7676  -34.3046
         0         0         0         0         0    2.3861   -0.8112
         0         0         0         0         0   -3.0933         0
         0         0         0         0         0         0   -3.0933
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
```

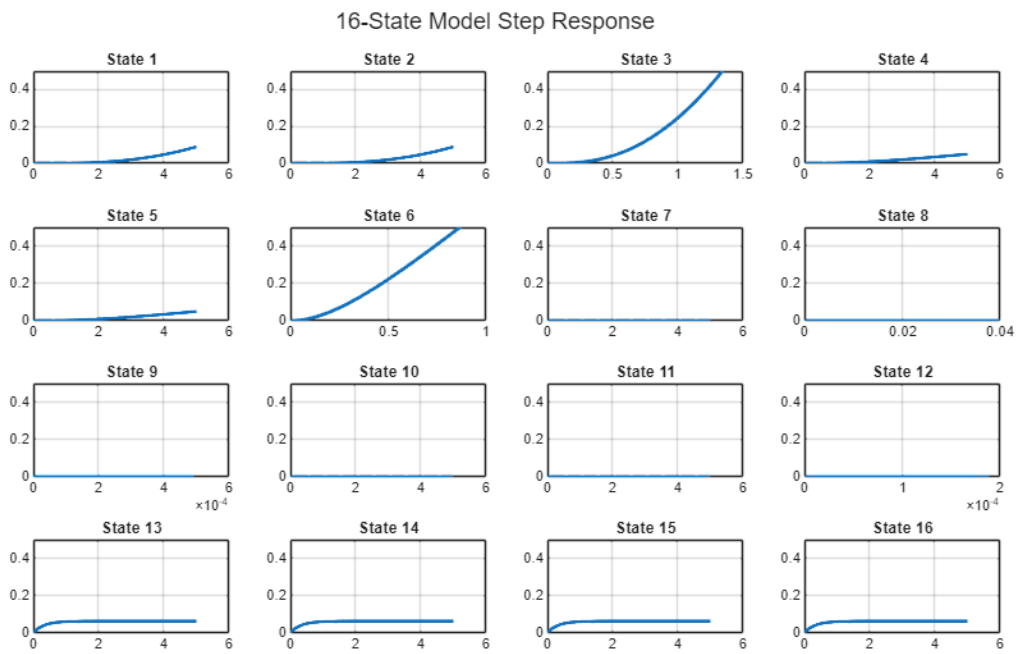
```
  Columns 15 through 16

         0         0
         0         0
         0         0
         0         0
         0         0
   -5.6040   -5.6040
         0         0
         0         0
         0         0
   38.0062   34.8319
  -38.5742   35.1111
    5.2097   -6.7846
         0         0
         0         0
   -3.0933         0
         0   -3.0933


B_2 =

     0     0     0     0
     0     0     0     0
     0     0     0     0
     0     0     0     0
     0     0     0     0
     0     0     0     0
     0     0     0     0
     0     0     0     0
     0     0     0     0
     0     0     0     0
     0     0     0     0
     0     0     0     0
     2     0     0     0
     0     2     0     0
     0     0     2     0
     0     0     0     2


C_2 =

  Columns 1 through 7

         0    1.0000         0         0         0         0         0
    1.0000         0         0         0         0         0         0
         0         0   -1.0000         0         0         0         0
         0         0         0         0    1.0000         0         0
         0         0         0    1.0000         0         0         0
         0         0         0         0         0   -1.0000         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0   57.2958
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
```

```
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0

  Columns 8 through 14

         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
   57.2958         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0  -57.2958         0         0         0         0         0
         0         0         0   57.2958         0         0         0
         0         0  -57.2958         0         0         0         0
         0         0         0         0  -57.2958         0         0
         0         0         0         0         0    1.4995         0
         0         0         0         0         0         0    1.4995
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0

  Columns 15 through 16

         0         0
         0         0
         0         0
         0         0
         0         0
         0         0
         0         0
         0         0
         0         0
         0         0
         0         0
         0         0
         0         0
         0         0
    1.4995         0
         0    1.4995


D_2 =

         0         0         0         0
         0         0         0         0
         0         0         0         0
         0         0         0         0
         0         0         0         0
         0         0         0         0
```

```
         0         0         0         0
         0         0         0         0
         0         0         0         0
         0         0         0         0
         0         0         0         0
         0         0         0         0
    0.0305         0         0         0
         0    0.0305         0         0
         0         0    0.0305         0
         0         0         0    0.0305
```

Step input for the 16-state model

```
sys_sim = ss(A_2, B_2, C_2, D_2);

% 10% increase from hover
u_step = u0_2 *1.1;

U_delta = (u_step - u0_2)';

t = 0:0.01:5;  % using the same sampling period .01s
U = repmat(U_delta, length(t), 1);

[y, tOut, x] = lsim(sys_sim, U, t);


for i = 1:16
    subplot(4, 4, i);
    plot(tOut, y(:,i), 'LineWidth', 2);
    ylim([0 0.5]);
    title(['State ' num2str(i)]);
    grid on;
end
sgtitle('16-State Model Step Response');
```



Albert Visualization.

```
% 1. Extract Raw States (Inertial NED & Euler Angles in rads)
% pn_sim = x(:, 1);
% pe_sim = x(:, 2);
% pd_sim = x(:, 3);
% phi_sim   = x(:, 7);
% theta_sim = x(:, 8);
% psi_sim   = x(:, 9);
%
% % 2. Define Drone Geometry for Plotting
% arm_length = L; % 0.046m based on your params
% d_geom = arm_length * (sqrt(2)/2);
% % Define the 4 motor positions in the body frame (X-configuration)
% motor_pos_body = [ d_geom, -d_geom, -d_geom,  d_geom;   % X coords
%                    d_geom,  d_geom, -d_geom, -d_geom;   % Y coords
%                    0,       0,       0,       0 ];      % Z coords
%
% % 3. Set up the Figure
% figure('Name', '3D Quadcopter Animation', 'Position', [100, 100, 800,
600]);
% hold on; grid on; axis equal;
% view([100, 45]);
%
% xlabel('East (m)');
% ylabel('North (m)');
% zlabel('Up (m)');
% title('Quadcopter 3D Animation');
%
```

```
% % Initialize graphics objects
% drone_lines = plot3(0,0,0, '-k', 'LineWidth', 1);
% motor_markers = plot3(0,0,0, 'ro', 'MarkerSize', 2, 'MarkerFaceColor',
'r');
% path_trace = plot3(pe_sim(1), pn_sim(1), -pd_sim(1), 'b--', 'LineWidth',
1.5);
%
% % Body Coordinate frame arrows (Red=X, Green=Y, Blue=Z)
% scale_arrow = arm_length * 1.5;
% q_x = quiver3(0,0,0, 1,0,0, scale_arrow, 'r', 'LineWidth', 1,
'MaxHeadSize', 0.5);
% q_y = quiver3(0,0,0, 0,1,0, scale_arrow, 'g', 'LineWidth', 1,
'MaxHeadSize', 0.5);
% q_z = quiver3(0,0,0, 0,0,1, scale_arrow, 'b', 'LineWidth', 1,
'MaxHeadSize', 0.5);
%
% % 4. Animation Loop
% disp('Starting 3D Animation...');
% for k = 1:length(tOut)
%     % Current position mapped to ENU (East, North, Up) for standard 3D
viewing
%     pos_enu = [pe_sim(k); pn_sim(k); -pd_sim(k)];
%
%     % Current attitude
%     rll = phi_sim(k);
%     ptc = theta_sim(k);
%     yw  = psi_sim(k);
%
%     % Rotation matrix: Body to Inertial (ZYX sequence matching your
dynamics)
%     Rx = [1 0 0; 0 cos(rll) -sin(rll); 0 sin(rll) cos(rll)];
%     Ry = [cos(ptc) 0 sin(ptc); 0 1 0; -sin(ptc) 0 cos(ptc)];
%     Rz = [cos(yw) -sin(yw) 0; sin(yw) cos(yw) 0; 0 0 1];
%     R_b2i = Rz * Ry * Rx;
%
%     % Transform matrix: NED to ENU for MATLAB plotting axes
%     R_ned2enu = [0 1 0; 1 0 0; 0 0 -1];
%
%     % Rotate geometry to Inertial (NED), convert to ENU, then translate
%     motor_pos_ned = R_b2i * motor_pos_body;
%     motor_pos_enu = R_ned2enu * motor_pos_ned + pos_enu;
%
%     % Update drone cross-frame (motor 1 to 3, motor 2 to 4)
%     X_arms = [motor_pos_enu(1,1), motor_pos_enu(1,3), NaN,
motor_pos_enu(1,2), motor_pos_enu(1,4)];
%     Y_arms = [motor_pos_enu(2,1), motor_pos_enu(2,3), NaN,
motor_pos_enu(2,2), motor_pos_enu(2,4)];
%     Z_arms = [motor_pos_enu(3,1), motor_pos_enu(3,3), NaN,
motor_pos_enu(3,2), motor_pos_enu(3,4)];
%
%     set(drone_lines, 'XData', X_arms, 'YData', Y_arms, 'ZData', Z_arms);
%     set(motor_markers, 'XData', motor_pos_enu(1,:), 'YData',
motor_pos_enu(2,:), 'ZData', motor_pos_enu(3,:));
%
```

```
%     % Update body coordinate axes vectors
%     axes_body = eye(3);
%     axes_ned = R_b2i * axes_body;
%     axes_enu = R_ned2enu * axes_ned;
%
%     set(q_x, 'XData', pos_enu(1), 'YData', pos_enu(2), 'ZData',
pos_enu(3), 'UData', axes_enu(1,1), 'VData', axes_enu(2,1), 'WData',
axes_enu(3,1));
%     set(q_y, 'XData', pos_enu(1), 'YData', pos_enu(2), 'ZData',
pos_enu(3), 'UData', axes_enu(1,2), 'VData', axes_enu(2,2), 'WData',
axes_enu(3,2));
%     set(q_z, 'XData', pos_enu(1), 'YData', pos_enu(2), 'ZData',
pos_enu(3), 'UData', axes_enu(1,3), 'VData', axes_enu(2,3), 'WData',
axes_enu(3,3));
%
%     % Update trajectory trace
%     set(path_trace, 'XData', pe_sim(1:k), 'YData', pn_sim(1:k), 'ZData',
-pd_sim(1:k));
%
%     % Tracking Camera: Moves the view window to follow the drone
%     % (Because the drone is 4cm wide but might fly several meters up)
%     xlim([-2, 2])
%     ylim([-2, 2])
%     zlim([0, 5])
%
%     drawnow;
% end
```

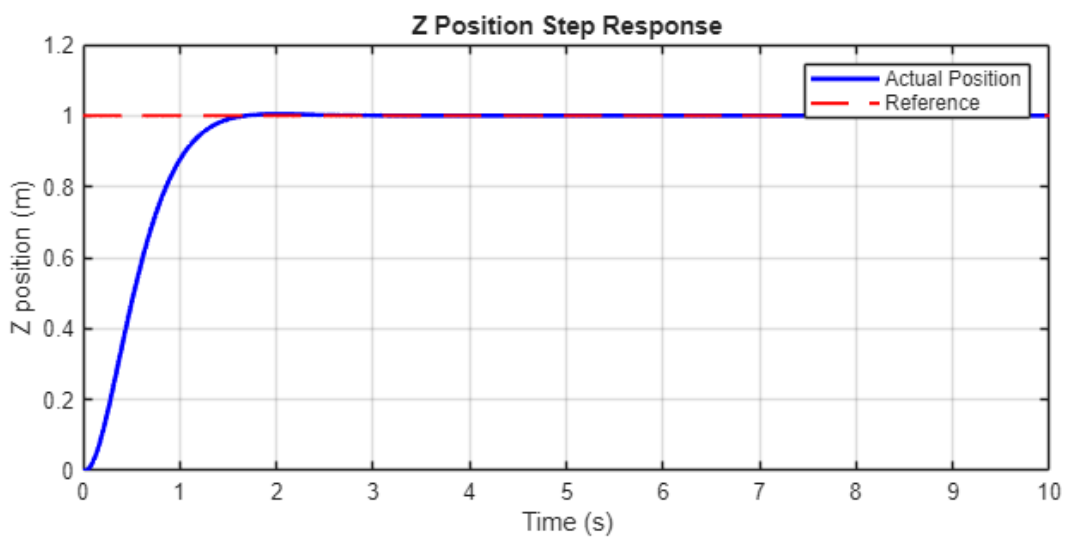
Altitude Control

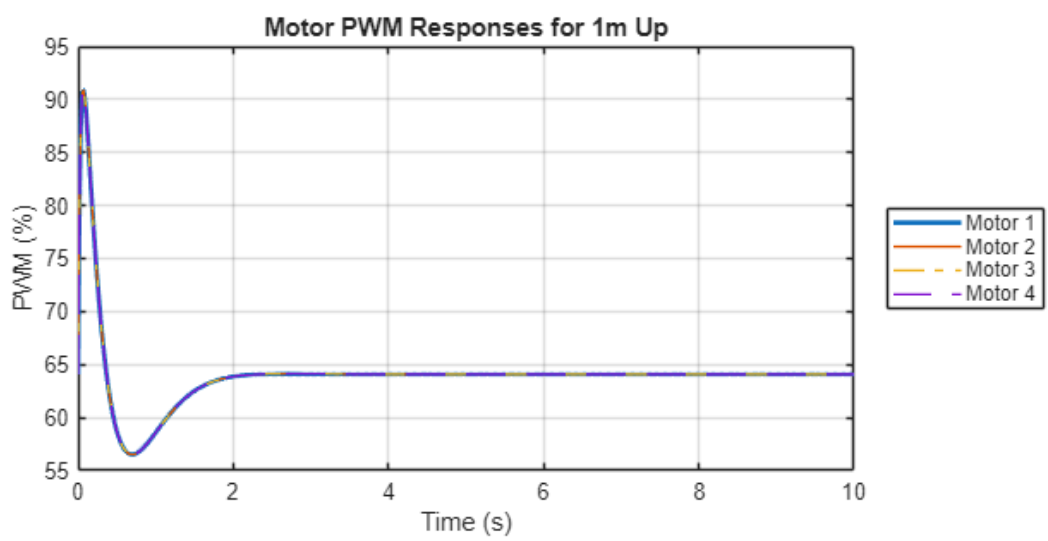
```
close all

% reduced - z, z_vel, and new z_error_int
A_z_aug = [0 1 0;
           0 0 0;
          -1 0 0]; % <-- should def become -1 (ENU)

B_z_aug = [0; 1/drone_mass_kg; 0];


Q_z = diag([10, 1, 100]); % penalizing state errors
R_z = 1; % penalizing control effort

K_z_fb = lqr(A_z_aug, B_z_aug, Q_z, R_z) % solve the risotto equation

% Transfer Function and Stability -----------------------------------
A_closed_loop = A_z_aug - B_z_aug*K_z_fb; % (A - BK)

B_cl_ref = [0; 0; 1];

sys_cl = ss(A_closed_loop, B_cl_ref, [1 0 0], 0);

figure
```

```
subplot(1,2,1); bode(sys_cl)
subplot(1,2,2); nyquist(sys_cl);
sgtitle('LQR Altitude Regulation')

% stability margin
G = tf(sys_cl);
figure
margin(G)




% Controller Simulation ----------------------------------------------

B_ref_z = [0; 0; 1];
sys_cl_z = ss(A_z_aug - B_z_aug*K_z_fb, B_ref_z, eye(3), zeros(3,1)); % (A -
BK)

t_sim = 0:0.01:10; % Slightly longer simulation to show settling
z_ref_step = ones(size(t_sim)); % 1 meter step command
[y_cl_z, t_cl, x_cl_z] = lsim(sys_cl_z, z_ref_step, t_sim);

figure('Name', 'Z Position Step Response & PWM', 'Position', [200, 150, 800,
800]);
subplot(2,1,1);
plot(t_cl, x_cl_z(:,1), 'b', 'LineWidth', 2); hold on;
plot(t_cl, z_ref_step, 'r--', 'LineWidth', 1.5);
title('Z Position Step Response');
xlabel('Time (s)'); ylabel('Z position (m)');
legend('Actual Position', 'Reference'); grid on;




% PWM behavior with Mixer ---------------------------------------------
F_cmd = -x_cl_z * K_z_fb';
F_hover = drone_mass_kg * 9.81; % We must maintain altitude!

pwm_hist_long = zeros(length(t_cl), 4);
for i = 1:length(t_cl)
    [pwm_pct, ~] = quad_mixer(F_hover + F_cmd(i), 0, 0, 0, A_thrust,
B_thrust, d, C_tau, F_min, F_max);
    pwm_hist_long(i, :) = pwm_pct';
end

subplot(2,1,2);
plot(t_cl, pwm_hist_long(:, 1), 'LineWidth', 2);
hold on
plot(t_cl, pwm_hist_long(:, 2), 'LineWidth', 1);
hold on
plot(t_cl, pwm_hist_long(:, 3), '-.', 'LineWidth', 1);
hold on
plot(t_cl, pwm_hist_long(:, 4), '--', 'LineWidth', 1);
```

```
title('Motor PWM Responses for 1m Up');
xlabel('Time (s)'); ylabel('PWM (%)');
legend('Motor 1', 'Motor 2', 'Motor 3', 'Motor 4', 'Location',
'EastOutside'); grid on;
```

```
K_z_fb =

    5.7890    1.1756  -10.0000
```



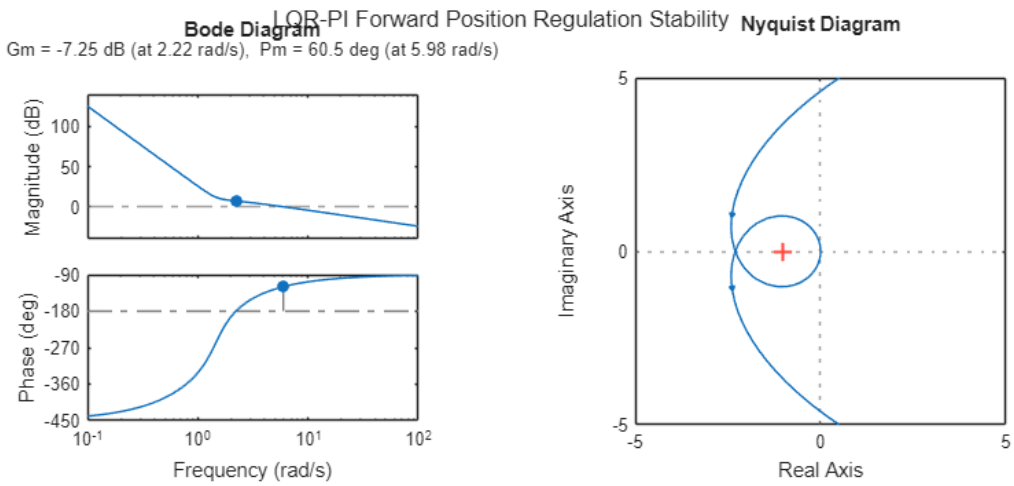


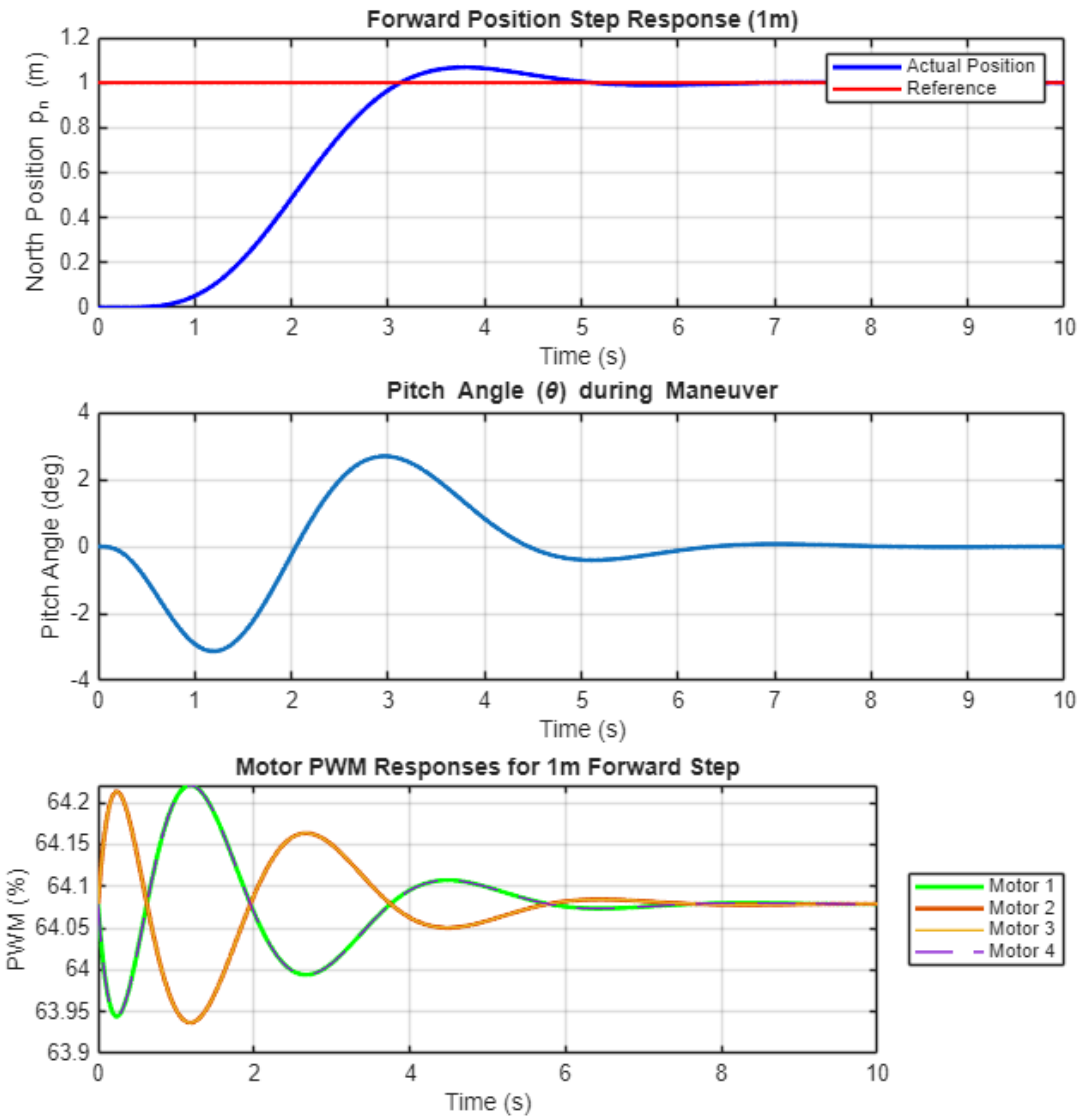
Forward Position Control

```
close all

A_pn = [0 1 0 0;
        0 0 -9.81 0;
        0 0 0 1;
        0 0 0 0];
B_pn = [0; 0; 0; 1/J(2,2)];




% state p, u, theta, q
A_pn_aug = [0 1 0 0 0;
            0 0 -9.81 0 0;
            0 0 0 1 0;
            0 0 0 0 0;
            -1 0 0 0 0]; % might need to change this row

B_pn_aug = [0; 0; 0; 1/J(2,2); 0];

Q_pn = diag([10, 10, 100, 10, 100]); % penalizing state errors
R_pn = 1e9; % penalizing control effort

K_pn = lqr(A_pn_aug, B_pn_aug, Q_pn, R_pn) % solve alfredo equation


%Stability Analysis ---------------------------------------------------
% 5 & 6. Open-Loop Transfer Function and Margins
% Break the loop at the plant input.
% P(s) = C_long * (sI - A_long)^-1 * B_long
% C(s) = K(1:4) * (sI - A_long + B_long*K(1:4))^-1 * B_long + K(5)/s
(Simplified approach below uses ss arrays)
sys_P = ss(A_pn, B_pn, eye(4), zeros(4,1));
L_ol_long = ss(A_pn_aug, B_pn_aug, K_pn, 0);

figure('Name', 'Forward Position Stability', 'Position', [150, 100, 1000,
400]);
subplot(1,2,1); margin(L_ol_long);
subplot(1,2,2); nyquist(L_ol_long);
xlim([-5, 5])
ylim([-5, 5])
sgtitle('LQR-PI Forward Position Regulation Stability');

[Gm_long, Pm_long, ~, ~] = margin(L_ol_long);
fprintf('Gain Margin: %.2f dB, Phase Margin: %.2f deg\n', 20*log10(Gm_long),
Pm_long);

%Closed-Loop Simulation ---------------------------------------------

B_ref_pn = [0; 0; 0; 0; 1]; % -1 for integral state
sys_cl_long = ss(A_pn_aug - B_pn_aug*K_pn, B_ref_pn, eye(5), zeros(5,1)); %
```

```
(A - BK)

t_sim = 0:0.01:10; % Slightly longer simulation to show settling
pn_ref_step = ones(size(t_sim)); % 1 meter step command
[y_cl_long, t_cl, x_cl_long] = lsim(sys_cl_long, pn_ref_step, t_sim);

figure('Name', 'Forward Position Step Response & PWM', 'Position', [200,
150, 800, 800]);
subplot(3,1,1);
plot(t_cl, x_cl_long(:,1), 'b', 'LineWidth', 2); hold on;
plot(t_cl, pn_ref_step, 'r', 'LineWidth', 1.5);
title('Forward Position Step Response (1m)');
xlabel('Time (s)'); ylabel('North Position p_n (m)');
legend('Actual Position', 'Reference'); grid on;

subplot(3,1,2);
plot(t_cl, x_cl_long(:,3)*(180/pi), 'LineWidth', 2);
title('Pitch Angle (\theta) during Maneuver');
xlabel('Time (s)'); ylabel('Pitch Angle (deg)'); grid on;


% PWM behavior with Mixer ------------------------------------------------
M_cmd = -x_cl_long * K_pn';
F_hover = drone_mass_kg * 9.81; % We must maintain altitude!

pwm_hist_long = zeros(length(t_cl), 4);
for i = 1:length(t_cl)
    % Mix hover thrust with commanded pitching moment. Roll and Yaw are 0.
    [pwm_pct, ~] = quad_mixer(F_hover, 0, M_cmd(i), 0, A_thrust, B_thrust,
d, C_tau, F_min, F_max);
    pwm_hist_long(i, :) = pwm_pct';
end

subplot(3,1,3);
plot(t_cl, pwm_hist_long(:,1), 'g-', 'LineWidth', 2); hold on
plot(t_cl, pwm_hist_long(:,2), 'LineWidth', 2); hold on
plot(t_cl, pwm_hist_long(:,3), 'LineWidth', 1); hold on
plot(t_cl, pwm_hist_long(:,4), '--','LineWidth', 1);


title('Motor PWM Responses for 1m Forward Step');
xlabel('Time (s)'); ylabel('PWM (%)');
legend('Motor 1', 'Motor 2', 'Motor 3', 'Motor 4', 'Location',
'EastOutside'); grid on;
```

```
K_pn =

   -0.0006   -0.0006    0.0031    0.0010    0.0003

Gain Margin: -7.25 dB, Phase Margin: 60.45 deg
```


LQR-PI Forward Position Regulation Stability
Bode Diagram
Gm = -7.25 dB (at 2.22 rad/s), Pm = 60.5 deg (at 5.98 rad/s)
Magnitude (dB)
100
50
0
Phase (deg)
-90
-180
-270
-360
-450
10^{-1}
10^{0}
10^{1}
10^{2}
Frequency (rad/s)
Nyquist Diagram
Imaginary Axis
5
0
-5
-5
0
5
Real Axis



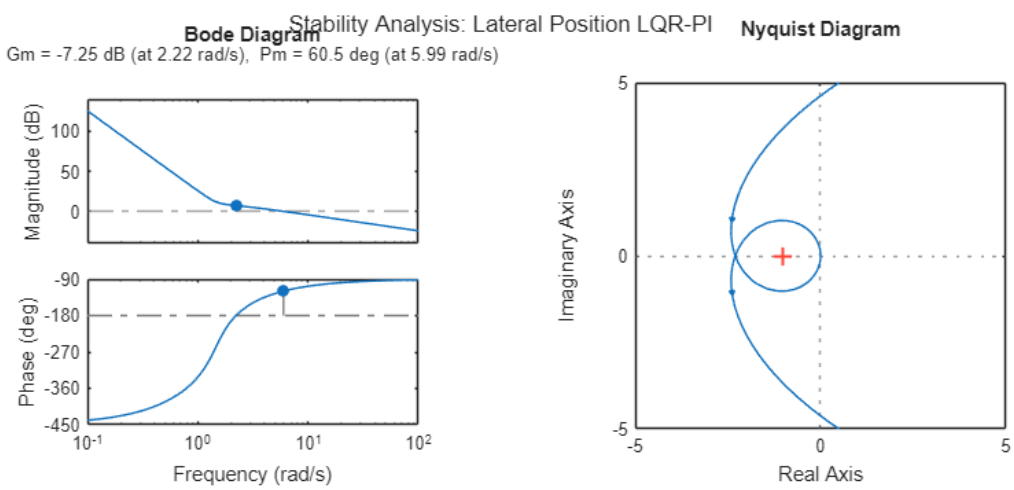
Lateral Position Control

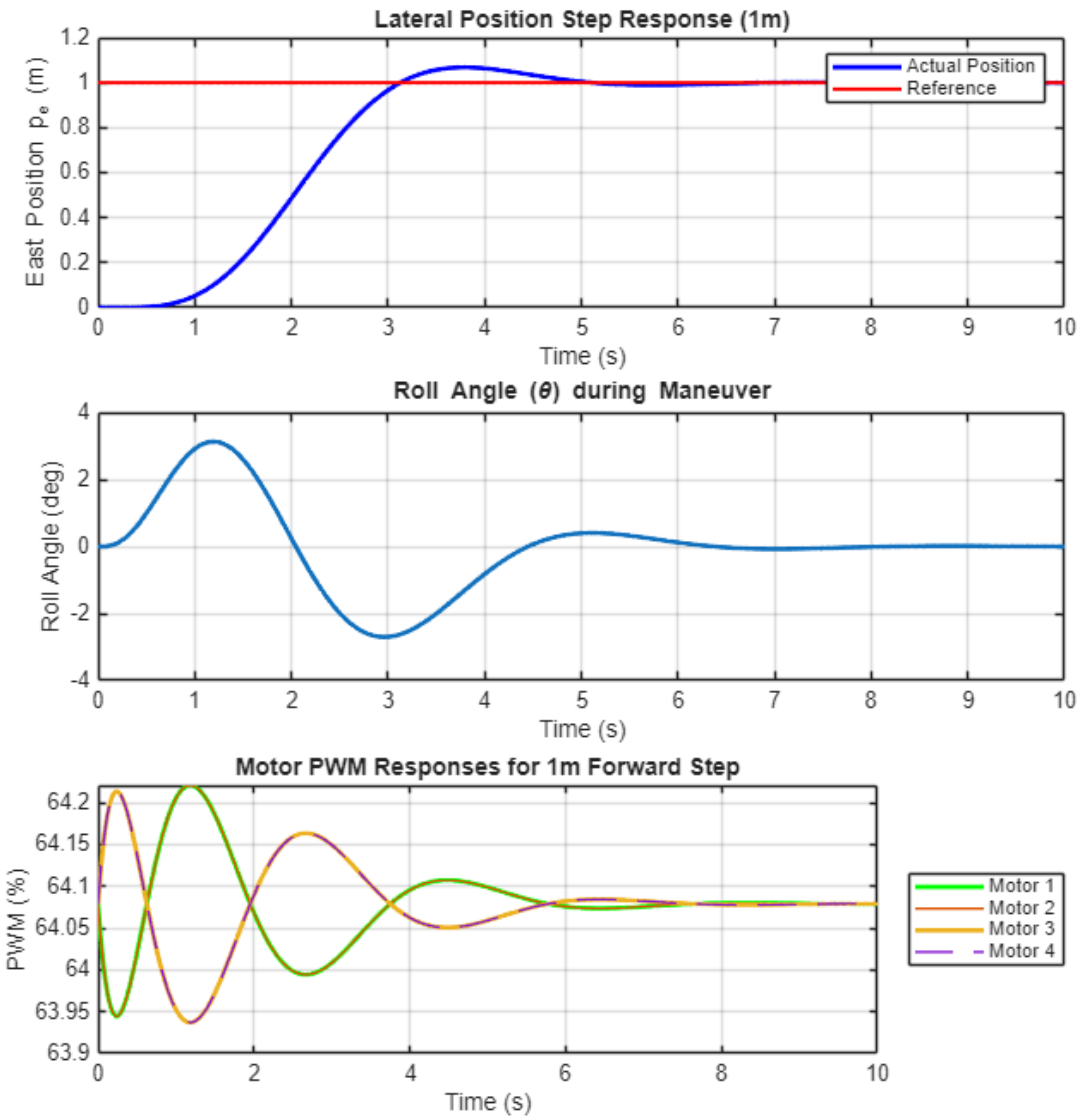
```
close all

% reduced state can be pe, v, phi, pv
A_pe = [0 1 0 0;
        0 0 9.81 0;
        0 0 0 1;
        0 0 0 0];

B_pe = [0; 0; 0; 1/J(1,1)];

[n_rows,n_col] = size(A_pe);
```

```
A_pe_aug = [A_pe zeros(n_rows,1);
            1, zeros(1,n_col)];


B_pe_aug = [B_pe; 0];

Q_pe = diag([10, 10, 100, 10, 100]); % penalizing state errors
R_pe = 1e9; % penalizing control effort

K_pe = lqr(A_pe_aug, B_pe_aug, Q_pe, R_pe)



%Transfer Functions and Stability -----------------------------------

%open loop TF
sys_L_ol = ss(A_pe_aug, B_pe_aug, K_pe, 0);

figure('Name', 'Lateral Position Stability', 'Position', [150, 100, 1000,
400]);
subplot(1,2,1); margin(sys_L_ol);
subplot(1,2,2); nyquist(sys_L_ol);
xlim([-5, 5])
ylim([-5, 5])
sgtitle('Stability Analysis: Lateral Position LQR-PI');

[Gm_long, Pm_long, ~, ~] = margin(sys_L_ol);
fprintf('Gain Margin: %.2f dB, Phase Margin: %.2f deg\n', 20*log10(Gm_long),
Pm_long);


%Closed-Loop Simulation -------------------------------------------

B_ref_pe = [0; 0; 0; 0; -1]; % -1 for integral state
sys_cl_lat = ss(A_pe_aug - B_pe_aug*K_pe, B_ref_pe, eye(5), zeros(5,1)); %
(A - BK) + B

t_sim = 0:0.01:10; % Slightly longer simulation to show settling
pe_ref_step = ones(size(t_sim)); % 1 meter step command
[y_cl_lat, t_cl, x_cl_lat] = lsim(sys_cl_lat, pe_ref_step, t_sim);

figure('Name', 'Lateral Position Step Response & PWM', 'Position', [200,
150, 800, 800]);
subplot(3,1,1);
plot(t_cl, x_cl_lat(:,1), 'b', 'LineWidth', 2); hold on;
plot(t_cl, pe_ref_step, 'r', 'LineWidth', 1.5);
title('Lateral Position Step Response (1m)');
xlabel('Time (s)'); ylabel('East Position p_e (m)');
legend('Actual Position', 'Reference'); grid on;

subplot(3,1,2);
plot(t_cl, x_cl_lat(:,3)*(180/pi), 'LineWidth', 2);
title('Roll Angle (\theta) during Maneuver');
xlabel('Time (s)'); ylabel('Roll Angle (deg)'); grid on;
```

```
% PWM behavior with Mixer ------------------------------------------------
L_cmd = -x_cl_lat * K_pe';
F_hover = drone_mass_kg * 9.81; % We must maintain altitude!

pwm_hist_long = zeros(length(t_cl), 4);
for i = 1:length(t_cl)

    [pwm_pct, ~] = quad_mixer(F_hover, L_cmd(i), 0, 0, A_thrust, B_thrust,
d, C_tau, F_min, F_max);
    pwm_hist_long(i, :) = pwm_pct';
end

subplot(3,1,3);
plot(t_cl, pwm_hist_long(:,1), 'g-', 'LineWidth', 2); hold on
plot(t_cl, pwm_hist_long(:,2), 'LineWidth', 1); hold on
plot(t_cl, pwm_hist_long(:,3), 'LineWidth', 2); hold on
plot(t_cl, pwm_hist_long(:,4), '--','LineWidth', 1);



title('Motor PWM Responses for 1m Forward Step');
xlabel('Time (s)'); ylabel('PWM (%)');
legend('Motor 1', 'Motor 2', 'Motor 3', 'Motor 4', 'Location',
'EastOutside'); grid on;
```

```
K_pe =

    0.0006    0.0006    0.0031    0.0010    0.0003

Gain Margin: -7.25 dB, Phase Margin: 60.45 deg
```





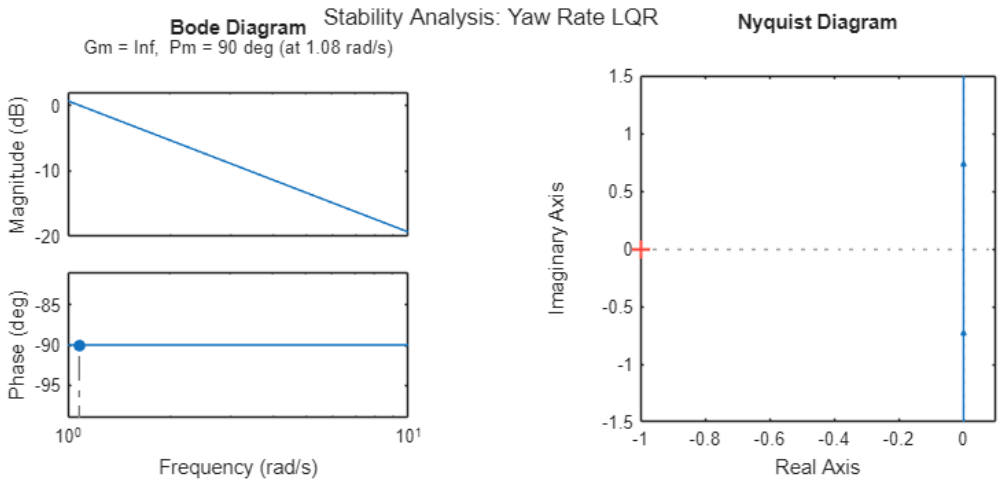
Problem 6: Yaw Rate Control (Damping)

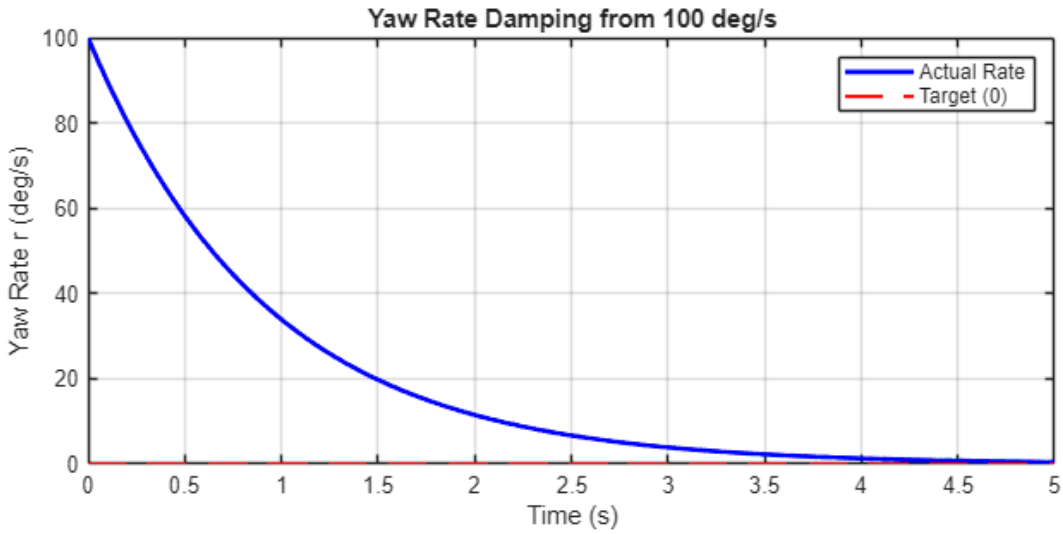
```
close all

% (1 & 2) Reduced state is just yaw rate (r). No integral state needed.
A_y = 0; % 1D system
B_y = 1/J(3,3);

Q_y = 1;
R_y = 1e7; % Penalize control effort to prevent aggressive motor saturation
K_y = lqr(A_y, B_y, Q_y, R_y);
fprintf('LQR Gain K_yaw: %.3f\n', K_y);
```

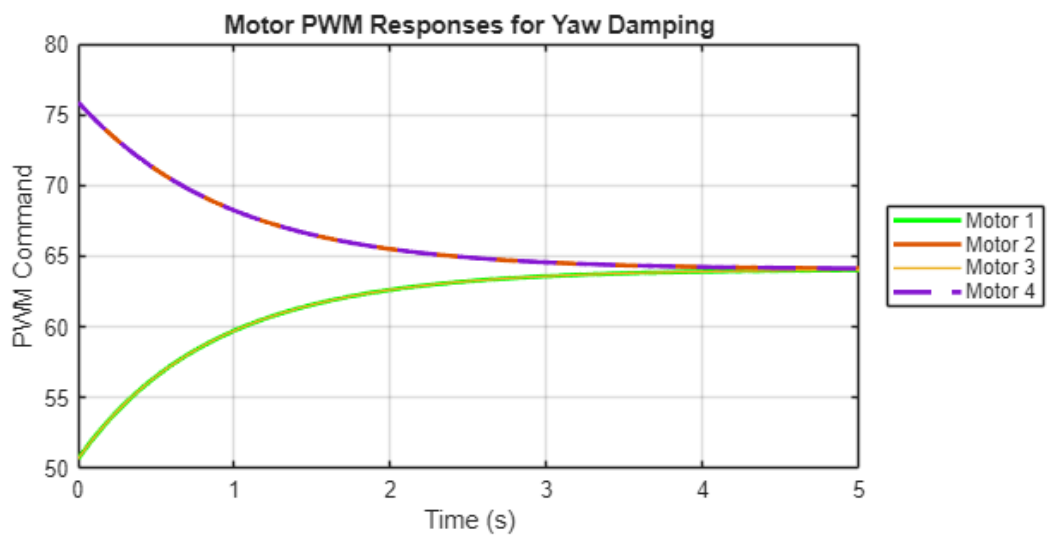
```
% (4, 5 & 6) Stability Analysis
sys_N_ol = ss(A_y, B_y, K_y, 0);

figure('Name', 'Yaw Stability', 'Position', [150, 100, 1000, 400]);
subplot(1,2,1); margin(sys_N_ol);
subplot(1,2,2); nyquist(sys_N_ol);
sgtitle('Stability Analysis: Yaw Rate LQR');

[Gm_yaw, Pm_yaw, ~, ~] = margin(sys_N_ol);
fprintf('Gain Margin: %.2f dB, Phase Margin: %.2f deg\n', 20*log10(Gm_yaw),
Pm_yaw);

% (7) Closed-Loop Simulation (Damping Initial Condition)
A_cl_yaw = A_y - B_y*K_y;
sys_cl_yaw = ss(A_cl_yaw, 0, 1, 0); % C=1, D=0

t_sim = 0:0.01:5; % 5 seconds is plenty for yaw
r0_yaw_rad = 100 * (pi/180); % Initial condition: 100 deg/s converted to
rad/s

[y_cl_yaw, t_cl, x_cl_yaw] = initial(sys_cl_yaw, r0_yaw_rad, t_sim);

figure('Name', 'Yaw Step Response & PWM', 'Position', [200, 150, 800, 800]);
subplot(2, 1, 1)
% Plot converted back to deg/s for easier reading
plot(t_cl, x_cl_yaw * (180/pi), 'b', 'LineWidth', 2); hold on;
plot(t_cl, zeros(size(t_cl)), 'r--', 'LineWidth', 1.5);
title('Yaw Rate Damping from 100 deg/s');
xlabel('Time (s)'); ylabel('Yaw Rate r (deg/s)');
legend('Actual Rate', 'Target (0)'); grid on;

% Estimate settling time using stepinfo logic adapted for initial conditions
info = stepinfo(x_cl_yaw, t_cl, 0, 'SettlingTimeThreshold', 0.05);
fprintf('Estimated time to damp yaw rate: %.2f seconds\n',
info.SettlingTime);

% (8) PWM behavior with Mixer
N_cmd = -x_cl_yaw * K_y; % Commanded YAWING moment
F_hover = drone_mass_kg * 9.81; % We must maintain altitude!

pwm_hist_yaw = zeros(length(t_cl), 4);
for i = 1:length(t_cl)
    % Mix hover thrust with commanded yawing moment (N). Roll and Pitch are
0.
    % Signature is (F, L, M, N, ...) -> N goes in the 4th slot!
    [pwm_pct, ~] = quad_mixer(F_hover, 0, 0, N_cmd(i), A_thrust, B_thrust,
d, C_tau, F_min, F_max);
    pwm_hist_yaw(i, :) = pwm_pct';
end

subplot(2,1,2);
plot(t_cl, pwm_hist_yaw(:,1), 'g-', 'LineWidth', 2); hold on
plot(t_cl, pwm_hist_yaw(:,2), 'LineWidth', 2); hold on
plot(t_cl, pwm_hist_yaw(:,3), 'LineWidth', 1); hold on
```

```
plot(t_cl, pwm_hist_yaw(:,4), '--','LineWidth', 2);
title('Motor PWM Responses for Yaw Damping');
xlabel('Time (s)'); ylabel('PWM Command');
legend('Motor 1', 'Motor 2', 'Motor 3', 'Motor 4', 'Location',
'EastOutside'); grid on;
```

```
LQR Gain K_yaw: 0.000
Gain Margin: Inf dB, Phase Margin: 90.00 deg
Estimated time to damp yaw rate: NaN seconds
```







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